# LaTeX Guide with Code Examples

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# Chapter 1

# Introduction to LaTeX

## 1.1 What is LaTeX?

LaTeX is a high-quality typesetting system widely used for scientific and technical documents. It is based on Donald Knuth's TeX and provides a structured way to write complex documents.

Some of the main features are:

- Automatic numbering of sections, figures, and tables.
- Bibliography and citation management.
- Mathematical typesetting.
- Professional layouts for reports, theses, and books.

## 1.2 Basic Structure of a Document

A typical LaTeX document has the following structure:

```
\documentclass{article}
\begin{document}
   Your content goes here...
\end{document}
```

The key parts are:

1. `\documentclass{}` – defines the type of document (e.g., article, report, book).
2. Preamble – where you load packages and set options.
3. Document body – content written between `\begin{document}` and `\end{document}`.

## 1.3 Levels of Sections

LaTeX provides a hierarchy of sectioning commands:

- `\chapter{}` – highest level (available in `report` and `book` classes).
- `\section{}` – major divisions.
- `\subsection{}` – subdivisions of sections.
- `\subsubsection{}` – smaller subdivisions.
- `\paragraph{}` and `\subparagraph{}` – lowest levels.

### 1.3.1 Example of Sectioning

Here is an example:

```
\chapter{Sample Chapter}
\section{First Section}
\subsection{A Subsection}
\subsubsection{A Subsubsection}
```

## 1.4 Text Formatting

You can apply formatting to text using commands:

- **Bold text** – `\textbf{...}`
- *Italic text* – `\textit{...}`
- Underlined text – `\underline{...}`
- `Verbatim text` – prints exactly as typed

## 1.5 Mathematics in LaTeX

Mathematics is one of the strongest features of LaTeX.
Inline math: $a^2 + b^2 = c^2$
Displayed math:

$$\int_0^\infty e^{-x}\,dx = 1$$

## 1.6 Lists in LaTeX

There are three main types of lists:

### 1.6.1 Itemized List

- First item
- Second item

### 1.6.2 Enumerated List

1. First item
2. Second item

### 1.6.3 Description List

**LaTeX** A document preparation system.

**TeX** The underlying typesetting engine.

## 1.7 Conclusion

This chapter introduced the basics of LaTeX, including its structure, sectioning, text formatting, mathematics, and lists. In the next chapters, we will explore figures, tables, and advanced features such as bibliographies.

# Chapter 2

# Latex code without compilation

This chapter contains how we can add latex code without compiling it directly.

## 2.1 Easiest way: verbatim

only thing is to do is : `\usepackage{verbatim}`

Listing 2.1: Verbatim

```
\begin{verbatim}
\documentclass{article}
\usepackage[utf8]{inputenc}

\title{My Document}
\author{Author Name}
\date{\today}

\begin{document}

\maketitle

\section{Introduction}
This is my first LaTeX document.

\end{document}
\end{verbatim}
```

Problem is very simple, no keyword coloring.

## 2.2 The lstlisting Environment :

The `\lstset` command configures the appearance of the code listings using the 'listings' package. You need to use packages : "listings" and "xcolor" xcolor in options for coloring.

### 2.2.1 Basic structure:

Listing 2.2: The basic structure of lstlisting

```
\begin{lstlisting}[caption={a caption}]
\documentclass{article}
\usepackage[utf8]{inputenc}
```

```latex

\title{My Document}
\author{Author Name}
\date{\today}

\begin{document}

\maketitle

\section{Introduction}
This is my first LaTeX document.
\end{document}
\ end{lstlisting}
% Here a space is added just because of the nexted lstlisting we do not need that
```

we need to first set up `\lstset` globally.

### 2.2.2 All the parameters of lstlisting:

Table 2.1: Complete list of listings package parameters

| Parameter | Values | Default | Description |
|---|---|---|---|
| `language` | Any supported language (C, C++, Python, Java, etc.) | none | Sets syntax highlighting language |
| `alsolanguage` | Another language | none | Adds another language's keywords |
| `deletekeywords` | {words} | none | Removes words from language's keywords |
| `morekeywords` | {words} | none | Adds extra keywords |
| `sensitive` | true / false | true | Case-sensitive keywords |
| `basicstyle` | Font commands | `\normalfont\ttfamily` | Base style for code |
| `keywordstyle` | Font/color | `\bfseries` | Style for keywords |
| `commentstyle` | Font/color | `\itshape` | Style for comments |
| `stringstyle` | Font/color | `\ttfamily` | Style for strings |
| `identifierstyle` | Font/color | same as basicstyle | Style for identifiers |
| `emph` | {words} | none | Emphasize specific words |
| `emphstyle` | Font/color | none | Style for emphasized words |
| `numbers` | none, left, right | none | Line numbering position |
| `stepnumber` | integer $\geq 1$ | 1 | Numbering interval |
| `firstnumber` | auto, integer | 1 | Starting line number |
| `numberstyle` | Font commands | `\tiny` | Style of numbers |
| `numbersep` | length | 10pt | Distance from numbers to code |
| `frame` | none, single, lines, shadowbox, topline, bottomline, leftline, rightline | none | Frame around code |
| `framerule` | length | 0.4pt | Frame thickness |
| `framesep` | length | 3pt | Distance between frame and code |
| `rulesep` | length | 2pt | Separation in shadowbox |

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| Parameter | Values | Default | Description |
|---|---|---|---|
| rulesepcolor | color | none | Color of rulesep |
| rulecolor | color | black | Frame color |
| backgroundcolor | \color{...} | none | Background color |
| aboveskip | length | \medskipamount | Vertical space above listing |
| belowskip | length | \medskipamount | Space below listing |
| abovecaptionskip | length | 10pt | Space above caption |
| belowcaptionskip | length | 0pt | Space below caption |
| xleftmargin | length | 0pt | Extra left margin |
| xrightmargin | length | 0pt | Extra right margin |
| framexleftmargin | length | 0pt | Left margin for frame only |
| framexrightmargin | length | 0pt | Right margin for frame only |
| breaklines | true / false | false | Enable automatic line breaking |
| breakatwhitespace | true / false | false | Break lines only at whitespace |
| breakindent | length | 20pt | Indentation after break |
| prebreak | text | \Righttorque | Symbol before broken line |
| postbreak | text | none | Symbol after line break |
| showspaces | true / false | false | Show visible spaces |
| showstringspaces | true / false | true | Show spaces in strings |
| showtabs | true / false | false | Show visible tabs |
| tabsize | integer | 8 | Number of spaces per tab |
| upquote | true / false | false | Straight quotes in code |
| literate | mapping | none | Replace sequences |
| caption | text | none | Caption text |
| captionpos | t / b | t | Caption placement (top/bottom) |
| label | text | none | Label for referencing |
| title | text | none | Alternative title |
| escapechar | char | none | Escape into LaTeX mode |
| escapeinside | {delim1}{delim2} | none | Escape with delimiters |
| mathescape | true / false | false | Enable math mode inside code |
| columns | fixed, flexible, fullflexible | flexible | Character spacing method |
| keepspaces | true / false | false | Preserve spaces as typed |
| inputencoding | encoding | system default | Input encoding |
| extendedchars | true / false | true | Allow extended characters |
| texcl | true / false | false | Treat % as LaTeX comment |
| inputpath | path | none | Default path for source files |
| alsoletter | characters | none | Treat additional characters as letters |
| alsodigit | characters | none | Treat additional characters as digits |
| classoffset | integer | 0 | Class offset for keyword classes |
| morecomment | [l][s][f]comment | none | Define additional comment styles |
| morestring | [b][d]delimiter | none | Define additional string delimiters |
| moredelim | *[s][...]... | none | Additional delimiters |
| moredelim | *[il][...]... | none | Additional inline delimiters |
| moredelim | *[is][...]... | none | Additional inline string delimiters |
| procnamekeys | language | none | Procedure name keywords |
| index | text | none | Index entry |

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| Parameter | Values | Default | Description |
|---|---|---|---|
| rangebeginprefix | text | none | Prefix for range begin |
| rangebeginsuffix | text | none | Suffix for range begin |
| rangeendprefix | text | none | Prefix for range end |
| rangeendsuffix | text | none | Suffix for range end |
| includerangemarker | true / false | true | Include range marker |
| linerange | start-end | none | Line range to include |
| linerangelabel | text | none | Label for line range |
| numberblanklines | true / false | true | Number blank lines |
| resetmargins | true / false | false | Reset margins |
| name | text | none | Name for the listing |
| formfeed | text | \newpage | Form feed character |
| gobble | integer | 0 | Number of characters to remove from each line |
| fancyvrb | true / false | false | Use fancyvrb package for verbatim |
| basewidth | length | 0.6em | Width of characters for alignment |
| fontadjust | true / false | true | Adjust font for alignment |
| selectchars | chars | none | Select specific characters |
| keywordsprefix | text | none | Prefix for keywords |
| keywordsuffix | text | none | Suffix for keywords |

## 2.3 Inline:

for inline command use \verb||